

E-Commerce Customer Segmentation using RFM Analysis

Project Overview:

The objective of this project was to perform customer segmentation on an e-commerce transaction dataset using the RFM (Recency, Frequency, Monetary) framework. The analysis aimed to identify distinct customer groups based on purchasing behaviour and translate those findings into business-oriented insights that could support retention, loyalty, and customer acquisition strategies.

The project consisted of four major stages:

1. Data preparation using PostgreSQL
2. RFM metric calculation and customer segmentation
3. Interactive dashboard development using Power BI
4. Business insight generation through exploratory analysis

Data Processing

Dataset

The project used the Online Retail transactional dataset containing customer purchase records.

Initial Dataset

- Total transaction records: **541,909**

The raw dataset contained several quality issues that required cleaning before analysis.

Data Cleaning

The following preprocessing steps were performed in PostgreSQL.

Missing Customer IDs

Rows without a CustomerID were removed because customer-level analysis cannot be performed without identifying the customer.

- Removed: **135,080** records

Cancelled Transactions

Invoices beginning with "C" represented cancelled or refunded transactions.

- Removed: **9,288** records

Invalid Quantities

Transactions with quantities less than or equal to zero were removed.

- Invalid records identified: **10,624**

Invalid Prices

Transactions with UnitPrice less than or equal to zero were removed.

- Invalid records identified: **2,521**

Final Clean Dataset

After completing all cleaning operations:

- Remaining transactions: **397,880**

A Revenue column was then created using:

Revenue = Quantity × UnitPrice

Total revenue across the cleaned dataset:

₹8,911,407.90

The latest transaction date in the dataset was:

09 December 2011

This date was used to calculate customer recency.

RFM Metric Generation

Each customer was evaluated using three behavioural metrics.

Recency

Number of days since the customer's last purchase.

Reference date:

10 December 2011

Frequency

Number of unique invoices generated by each customer.

Monetary

Total amount spent by each customer across all valid transactions.

Customer Base

After grouping transactions by CustomerID:

- Total customers: **4,338**

RFM Scoring

Customers were divided into five equal groups using SQL window functions.

Scores ranged from:

- 1 = Lowest
- 5 = Highest

Separate scores were assigned for:

- Recency
- Frequency
- Monetary

The three scores were then combined into a three-digit RFM score.

Example:

- 555
- 442
- 111

Customer Segmentation

Customers were classified into six behavioural groups.

Segment	Customer s
Champions	605
Loyal Customers	829
Potential Loyalists	484
At Risk	654
Hibernating	1,082
Others	684

Revenue generated by each segment:

Segment	Revenue
Champions	₹5,236,938.27
Loyal Customers	₹1,442,466.40
At Risk	₹901,325.71
Potential Loyalists	₹504,702.67
Hibernating	₹450,707.10
Others	₹375,267.75

Power BI Dashboard

An interactive dashboard was developed to explore customer behaviour dynamically.

The dashboard contains:

- KPI Cards
 - Total Customers
 - Champions Count
 - Total Revenue
 - At Risk Count
- Revenue by Segment
- Customer Segment Distribution
- RFM Heatmap
- Customer Detail Table
- Segment Slicer

The dashboard supports cross-filtering between visuals, allowing individual customer segments and RFM score combinations to be explored interactively.

Observations and Insights

Observation Summary:

The provided observations provide us with 3 key decision-making factors:

- a) The First Priority customers (FCs) contribute to about 50% of the total revenue, despite being only 10% in number. Efforts and budget towards customer care, marketing and sales can be focused on this target demographic.
- b) About 15% of the customers come under the At Risk segment. Regulating customer satisfaction is of high priority to increase purchase frequency for this group.
- c) The Hibernating customers comprise of 24% in number, but generate the least in revenue. But about 10% of them can become Potential Loyalists since they have a high recency score. Maintaining and increasing this number is important for competence and marketing purposes.

Observation 1: Dashboard Overview

The dashboard shows that the dataset has a total of 4,338 unique customers, with 605 Champions and 654 At Risk, and the total revenue is estimated at 8.91 Million Rupees. The revenue-per-segment Bar chart shows that the Champions contribute to about 5.2 Million Rupees, followed by the Loyal Customers at about 1.4 Million Rupees, which are then followed by the At

Risk, the Potential Loyalists, the Hibernating, and the Others, whose revenue amounts to 0.9, 0.5, 0.5, and 0.4 Million Rupees, respectively.

The customer-per-segment Donut chart shows that the Hibernating is the most dominant in number of customers(about 24.5%), compared to the rest of the segments. The Loyal Customers follow up with 19.1%. The remaining segments range from 11.6% to 15.7%.

The rfm_score heatmap shows that the most dense part of the total revenue comes from those customers with a recency score, frequency score and monetary score of 5, 5 and 5, respectively. This group of customers can be classified as First Priority(FP) customers.

Observation 2: The FP Customers Analysis

The charts show that the most valuable set of customers with an RFM score of 555 bring in a total of 4.04 Million Rupees, which is almost 50% of the estimated total revenue. Consisting of just 437 total customers in number, they come from mainly two segments- the Champions(8.02%), and the Loyal Customers(2.05%).

348 out of the 437 FP customers come from the Champions segment(about 58% out of the total no. of Champions), bringing in 3.91 Million Rupees in value. The remaining 89 customers come from the Loyal Customers segment(about 10.8% out of the total no. of LCs), bringing in 132.64 Thousand Rupees in value.

Observation 3: Segment-wise General Analysis

1. **Champions:** 605 total customers, 5.24 Million Rupees in revenue. 348 out of the total champions (FPs) bring in 3.91 Million Rupees in revenue.
2. **Loyal Customers:** 829 total customers, 1.44 Million Rupees in revenue. 89 out of the total LCs (FPs) bring in 132.64 Thousand Rupees in revenue.
3. **At Risk:** 654 total customers, generating 901.33 Thousand Rupees in revenue. 53 out of 654 (8.1%) bring in 154.98 Thousand Rupees in revenue.
4. **Potential Loyalists:** 484 total customers, generating 504.7 Thousand Rupees in revenue.
5. **Hibernating:** 1082 total customers, generating 450.71 Thousand Rupees in revenue. 357 out of 1082 (8.23%) bring in 186.12 Thousand Rupees in revenue.
6. **Others:** 684 total customers, generating 375.27 Thousand Rupees in revenue. 204 out of 684 (29.82%) bring in 195.27 Thousand Rupees in revenue.

Conclusion

The completed RFM analysis successfully transformed raw transactional data into actionable customer segments through systematic data preparation, behavioural scoring, and interactive visualization. Starting with 541,909 transaction records, the dataset was cleaned to 397,880 valid transactions before calculating customer-level Recency, Frequency, and Monetary metrics for 4,338 unique customers.

The segmentation process revealed that customer value is not evenly distributed. Although Champions account for only 605 customers (approximately 13.95% of the customer base), they generate ₹5.24M of the total ₹8.91M revenue, representing approximately 58.8% of total revenue. This concentration illustrates the importance of retaining high-value customers while developing targeted strategies for customers in the At Risk, Loyal Customers, Potential Loyalists, Hibernating, and Others segments.

The Power BI dashboard complements the SQL analysis by enabling interactive exploration through segment filters, RFM score cross-filtering, revenue comparisons, and customer-level drill-downs. Together, the SQL pipeline and dashboard provide a complete analytical workflow—from raw transaction data to business-ready insights—demonstrating how RFM analysis can support customer retention, loyalty initiatives, and marketing decision-making.

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